INCH-POUND
MIL-PRF-1/267F
20 March 2007
SUPERSEDING
MIL-E-1/267E
18 August 1971

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 3WP1 AND 3WP2

Inactive for new design
after 7 March 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

PIN CONNECTIONS AND DIMENSIONS: See figure 1.

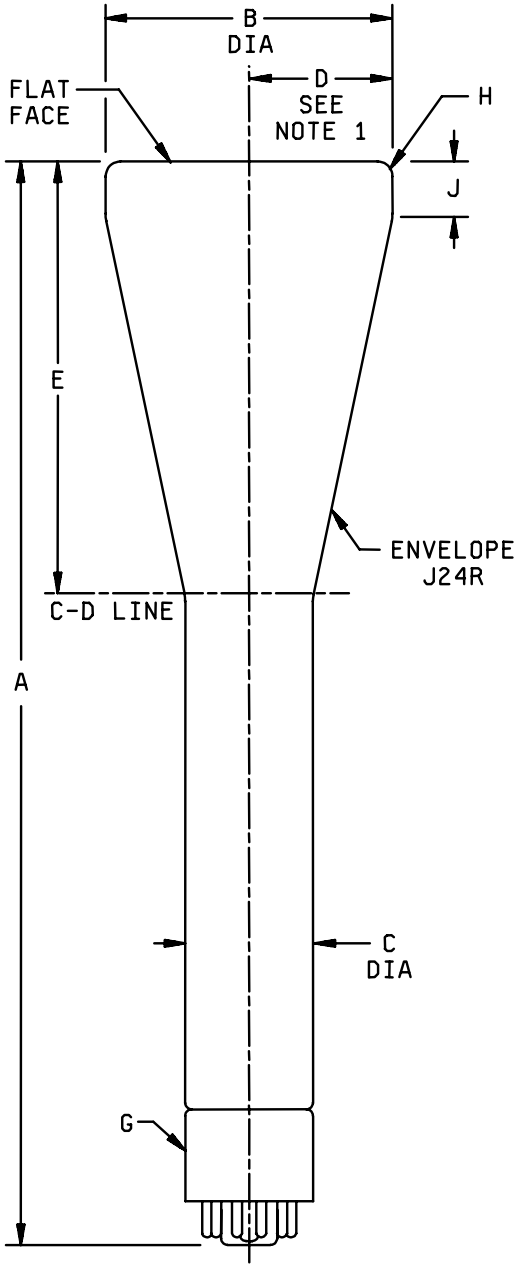
ABSOLUTE RATINGS:

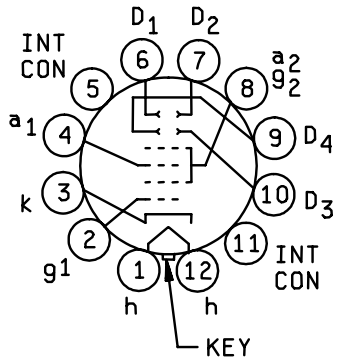
Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Ehk	Alt
Unit:	V	V dc	V dc	V dc	V dc	MegΩ	MegΩ	V dc	ft
Maximum:	6.9	0, -200	550	1,100	2,750	1.5	1.0	±180	30,000
Minimum:	5.7	---	---	---	1,000	---	---	---	---
Test conditions:	6.3	Adjust	---	Focus	1,500	---	---	---	---

GENERAL:

Qualification - Required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:07Z
Check the source to verify that this is the current version before use.





BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	2.938 (74.63)	3.063 (77.80)
G	Base: B12-43 or B10-75	
Conformance inspection, part 2		
A	11.375 (288.93)	11.625 (295.28)
C	1.313 (33.35)	1.438 (3;6.53)
D	1.375 (34.93) R	---
Reference dimensions (see note 2)		
E	4.563 (115.90)	
H	0.063 (1.60) R	
J	0.563 (14.30)	

NOTES:
1. Useful screen radius.
2. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3WP1 and 3WP2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:07Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and Inspection.

MIL-STD-1311 method	Requirement or test	Type	Conditions	Acceptance level (see note 3)	Symbol	Limits		Unit
						Min	Max	
	Qualification inspection							
1216	Base material insulating quality	Both	Zone 5, or higher	---	---	---	---	---
1141	Pressure (implosion)	Both		---	---	---	---	---
1002	Barometric pressure, reduced	Both	225 mmHg	---	---	---	---	---
5111	Vibration	Both		---	Width	---	1	mm
1331	Direct-interelectrode capacitance	Both	g1 to all	---	Cg1	---	8.7	pF
			k to all	---	Ck	---	5.7	pF
			D1 to D2	---	C1D2	---	3.3	pF
			D3 to D4	---	C3D4	---	2.0	pF
			D1 to all, except D2	---	CD1	---	7;.2	pF
			D2 to all, except D1	---	CD2	---	7.2	pF
			D3 to all, except D4	---	CD3	---	4.8	pF
			D4 to all, except D3	---	CD4	---	4.8	pF
5101	Neck and bulb alignment (electrostatic types)	Both		---	Dia.	---	1.63	Inch
5216	Cathode illumination	Both		---	---	---	---	---
5248	Deflection-factor uniformity	Both		---	---	---	2	%
	Conformance inspection, part 1							
5106	Bulb, screen, and faceplate quality	Both		---	---	---	---	---
5201	Voltage breakdown	Both		---	---	---	---	---
5201	Voltage breakdown, (electrostatic types)	Both		---	---	---	---	---
5206	Gas "cross"	P1	Light = 7 fL (see note 1)	---	---	---	---	---
5206	Gas "cross"	P2	Light = 2 fL (see note 1)	---	---	---	---	---
5221	Light output	P1	See note 1	---	Light	7	---	fL
5221	Light output	P2	See note 1	---	Light	2	---	fL
5231	Spot position (electrostatic deflection)	Both		---	---	---	10	mm
5231	Spot displacement (leakage)	Both		---	Displ	---	7	mm
5241	Grid cutoff voltage	Both		---	Eco	-45	-75	V dc
5251	Grid No. 1 leakage current	Both		---	---	---	---	---
5251	Anode No. 2 leakage current	Both		---	---	---	---	---

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:07Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and Inspection - Continued.

MIL-STD-1311 method	Requirement or test	Type	Conditions	Acceptance level (see note 3)	Symbol	Limits		Unit
	Conformance inspection, part 1 - Continued							
---	Useful scan	Both	Measured from tube face center; Focused trace					
			1D2 scan	---	---	2.50	---	Inch
			3D4 scan	---	---	2.25	---	Inch
5103	Pattern distortion	Both	See note 2	---	---	---	---	---
	Conformance inspection, part 2							
1301	Heater current	Both		---	If	540	660	mA
5201	Electrode current (cathode)	P1	Light = 7 fL	---	Ik	---	1,000	μA dc
5201	Electrode current (cathode)	P2	Light = 2 fL	---	Ik	---	1,000	μA dc
5201	Electrode current (anode No. 1)	Both	Ec1 = 0	---	Ib1	-15	10	μA dc
5101	Base alignment (electrostatic types)	Both	+1D2; pin No. 3	---	---	---	---	---
5101	Neck and base alignment (electrostatic types)	Both		---	---	---	---	---
5101	Angle between traces	Both		---	---	89	91	De-grees
5216	Stray light emission (conventional types)	Both	Eb2 = 2,750 V dc	---	---	---	---	---
5223	Modulation	P1	Light = 7 fL	---	ΔEc	---	50	V dc
5223	Modulation	P2	Light = 2 fL	---	ΔEc	---	45	V dc
5226	Line width A (electrostatic deflection)	P1	Light = 7 fL	---	Width	---	0.65	mm
5226	Line width A (electrostatic deflection)	P2	Light = 2 fL	---	Width	---	0.65	mm
5226	Line width B (electrostatic deflection)	P1	Light = 7 fL	---	Width	---	0.75	mm
5226	Line width B (electrostatic deflection)	P2	Light = 2 fL	---	Width	---	0.75	mm
5246	Focusing voltage at modulation condition	Both		---	Eb1	247	465	V dc
5248	Deflection factor	Both	1D2	---	DF	62	76	V dc/ in.
5248	Deflection factor	Both	3D4	---	DF	43	52	V dc/ in.

See notes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:07Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and Inspection - Continued.

MIL-STD-1311 method	Requirement or test	Type	Conditions	Acceptance level (see note 3)	Symbol	Limits		Unit
	Conformance inspection, part 2 - Continued							
5251	Heater-cathode leakage current	Both		---	---	---	---	---
1111	Base pin solder depth	Both		---	---	---	---	---
1101	Secureness of base, cap, or insert	Both		---	---	---	---	---
1105	Permanence of marking	Both		---	---	---	---	---
	Conformance inspection, part 3							
---	Life-test	Both	Group C; t = 500 hours minimum; Eb2 = 2,750 V dc	---	---	---	---	---
		P1	Light = 7 fL	---	---	---	---	---
		P2	Light = 2 fL	---	---	---	---	---
---	Life-test end points	Both	Line width A	---	Width	---	0.65	mm
			Line width B	---	Width	---	0.75	mm
			Heater-cathode leakage current	---	---	---	---	---
			Grid No. 1 leakage current	---	---	---	---	---
			Grid No. 2 leakage current	---	---	---	---	---
			Anode No. 1 leakage current	---	---	---	---	---
			Stray light emission	---	---	---	---	---
		P1	Light = 4.5 fL	---	---	---	---	---
			Modulation	---	ΔEc	---	50	V dc
		P2	Light = 1.5 fL	---	---	---	---	---
			Modulation	---	ΔEc	---	45	V dc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. With a raster pattern adjusted so that its widest point just touches the sides of a 2.05-inch (52.07 mm) square, no point on the raster sides shall be within an inscribed 1.95-inch (49.53 mm) square.
3. This specification sheet uses accept on zero sampling in accordance with MIL-PRF-1, table III.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:07Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-2007-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:07Z
Check the source to verify that this is the current version before use.